Drug Repurposing Hypothesis Assessment: Ensifentrine in Idiopathic Pulmonary Fibrosis

Summary

The hypothesis holds that **ensifentrine** — an inhaled dual inhibitor of phosphodiesterases 3 and 4 (PDE3/PDE4), currently FDA-approved only for COPD maintenance — would benefit patients with **idiopathic pulmonary fibrosis (IPF)** by interrupting fibroblast activation, myofibroblast differentiation, and profibrotic inflammation, the same disease mechanism that oral PDE4B-preferential inhibition with nerandomilast is reported to slow. After five iterations, 9 confirmed findings, and review of 73 papers, the verdict is that the hypothesis is **mechanistically plausible but speculative and currently unproven**. Evidence both for and against was weighed; the weight of evidence supports the *pathway*, not the *molecule*.

The strongest argument in favor is that the cAMP/phosphodiesterase pathway is now a **clinically validated antifibrotic target in IPF** — but it was validated by a *different drug*. The Phase 3 FIBRONEER-IPF trial of oral nerandomilast (a PDE4B-preferential inhibitor) showed a statistically significant slowing of forced vital capacity (FVC) decline (PMID: 40387033). The pathway acts across exactly the cellular compartments central to IPF (fibroblasts, macrophages, alveolar epithelium, vasculature), PDE4 inhibition is antifibrotic in preclinical lung-fibrosis models, and — importantly for a dual inhibitor — PDE3 inhibition is independently antifibrotic through the same cAMP-elevating mechanism. The inhaled route is a genuine strategic advantage, potentially sparing patients the gastrointestinal toxicity that limits oral PDE4 inhibitors.

The arguments against are substantial and should temper enthusiasm. **Ensifentrine itself has zero fibrosis data at any level** — no in vitro, no animal, no human studies in IPF or any fibrotic disease; its entire evidence base is COPD. The target is **not genetically implicated** in IPF (the disease is driven by MUC5B, telomere, and surfactant biology, not cAMP machinery), meaning this is a downstream effector target rather than a disease driver. Inhaled delivery to dense, mucus-obstructed, ECM-stiffened fibrotic lung is a real and specific pharmacokinetic barrier. The PDE3 arm carries a vasodilation signal that was catastrophic for *systemic*

vasodilators in fibrotic ILD (though inhaled delivery plausibly mitigates this). And even if everything works, the mechanism can at best *slow* decline, not reverse established fibrosis. The net assessment is a reasonable repurposing idea that warrants a human-tissue proof-of-mechanism study before any large FVC-endpoint trial — **low confidence, worth testing, not yet justified for a pivotal trial.**

Key Findings

Finding 1 — The cAMP/PDE pathway is a clinically validated antifibrotic target in IPF (but validated by nerandomilast, not ensifentrine)

The single most important piece of evidence supporting the hypothesis's *pathway* is the Phase 3 FIBRONEER-IPF trial (NCT05321069, n=1177) of oral nerandomilast, a PDE4B-preferential inhibitor. The adjusted mean FVC change at week 52 was −114.7 mL on nerandomilast 18 mg BID versus −183.5 mL on placebo — an adjusted difference of 68.8 mL (95% CI 30.3–107.4; P<0.001). The 9-mg dose showed a smaller but still significant difference of 44.9 mL (95% CI 6.4–83.3; P=0.02). Notably, 77.7% of patients were on background nintedanib or pirfenidone, so this benefit was *additive* to standard of care. The most frequent adverse event was diarrhea.

> *"The adjusted difference between the nerandomilast 18-mg group and the placebo group was 68.8 ml (95% confidence interval [CI], 30.3 to 107.4; P<0.001)"* — PMID: 40387033

Interpretation (established): PDE4B inhibition slows FVC decline in IPF, establishing the cAMP/PDE pathway as a legitimate antifibrotic target. **Inference (ours):** This validates the *pathway* ensifentrine engages, but a 69 mL benefit over 52 weeks is a partial slowing, not stabilization — it defines both the opportunity and the ceiling.

Finding 2 — Ensifentrine's clinical evidence is confined to COPD; no IPF studies exist

Systematic PubMed searches on 2026-11-07 for "ensifentrine idiopathic pulmonary fibrosis" and "ensifentrine interstitial lung disease" returned **zero papers**. Ensifentrine's pivotal evidence is entirely in COPD: the ENHANCE-1 (NCT04535986) and ENHANCE-2 (NCT04542057) trials supported its FDA approval for COPD maintenance. No registered IPF trial of ensifentrine was identified.

> *"Ensifentrine, recently approved by the FDA for chronic obstructive pulmonary disease (COPD) maintenance treatment, is a novel inhaled therapy with a dual mechanism of action targeting phosphodiesterase (PDE)3 and PDE4."* — PMID: 40331465

Interpretation (established): The drug's entire clinical identity is COPD. **Inference (ours):** Any claim of IPF benefit is pure extrapolation from a related but distinct disease and from a different (oral, PDE4B-preferential) molecule.

Finding 3 — The PDE4/cAMP pathway is mechanistically active across IPF effector cells

Review evidence establishes that PDE4 enzymes are highly expressed in the lung and that, via cAMP modulation, PDE4 inhibition exerts effects on TGF-β-dependent and -independent signaling across immune cells, macrophages, alveolar epithelial cells, fibroblasts, and the pulmonary vasculature. A PDE4 inhibitor (DC591017) reduced collagen deposition comparably to nintedanib in bleomycin mice by directly inhibiting fibroblast activation and epithelial–mesenchymal transition (EMT).

> *"These effects extend across immune cells, macrophages, alveolar epithelial cells, fibroblasts, and the pulmonary vasculature."* — PMID: 42768757
>
> *"DC591017 exerted anti-fibrotic effects by directly inhibiting fibroblasts activation and migration, and suppressing epithelial-mesenchymal transition (EMT)"* — PMID: 41771338

Interpretation (established): The pathway acts precisely on the cellular compartments the hypothesis invokes. This directly addresses Question 1 (is the mechanism active in the disease): yes, at the tissue/cellular level.

Finding 4 — PDE3 inhibition (ensifentrine's *second* target) is independently antifibrotic in lung models

A critical point distinguishing ensifentrine from nerandomilast is its PDE3 activity. Cilostazol, a selective PDE3 inhibitor, is being repurposed for pulmonary fibrosis: network pharmacology flagged PDE3 as a key antifibrotic target, and bleomycin-rat validation showed anti-inflammatory/antifibrotic effects. Cilostazol is antifibrotic across multiple organ-fibrosis models by raising cAMP and blocking myofibroblast conversion, and cAMP elevation is intrinsically antifibrotic in lung fibroblasts.

> *"Network pharmacology analysis identified 10 potential targets of cilostazol, of which eight emerged as key network regulators: phosphodiesterase 3 (PDE3)"* — PMID: 42207281
>
> *"The second messenger molecule cyclic 3',5'-adenosine monophosphate (cAMP) has antifibrotic properties"* — PMID: 31884570

Interpretation (ours): The PDE3 arm adds an independent antifibrotic rationale, so dual PDE3/4 inhibition could in principle elevate cAMP more fully than PDE4B inhibition alone. This is the strongest *pharmacological* argument that ensifentrine might do something nerandomilast does — or more.

Finding 5 — The target is NOT genetically implicated in IPF — it is a downstream effector, not a disease driver

IPF genetic risk is dominated by the MUC5B promoter variant (rs35705950) and rare variants in telomere-maintenance and surfactant genes, plus cell-senescence and host-defense loci. No GWAS or linkage signal implicates PDE3, PDE4, or the cAMP machinery.

> *"family-based studies have led to the identification of rare genetic variants in genes related to surfactant function and telomere biology"* — PMID: 25837031

Confirmatory resequencing implicates host-defense and cell-senescence genes (PMID: 31034279).

Interpretation (ours): Human genetics — the most causally rigorous line of evidence — does not point to this pathway. The pathway is a therapeutically useful downstream node (as nerandomilast proves), but it is not where the disease *originates*. This is important because effector-target drugs tend to modify, not cure, disease.

Finding 6 — Inhaled delivery to fibrotic lung is a real mechanistic barrier (the "traffic-jam" effect)

In IPF, mucus hypersecretion and ECM stiffening form a physical obstruction that restricts inhaled-drug penetration and retention. Specialized dual-enzyme nanoparticles were required to raise aerosol deposition to 86.5%, and aerosolized compounds show heterogeneous intrapulmonary distribution in bleomycin fibrosis.

> *"pathological mucus accumulation and ECM stiffening jointly form a"* — PMID: 42359544

Heterogeneous aerosol distribution in fibrotic lung is independently documented (PMID: 37843890).

Interpretation (ours): Ensifentrine's inhaled route — an advantage for systemic tolerability — may be a *liability* for reaching the fibrotic lesions themselves. Nerandomilast is oral and reaches lung via the circulation, bypassing this barrier. This is a central, drug-specific reason the hypothesis might fail.

Finding 7 — Pulmonary vasodilation is a double-edged safety flag; inhaled delivery mitigates but does not eliminate it

Ensifentrine's PDE3 activity is vasodilatory. Systemic PAH-targeted vasodilators have been *harmful* in fibrotic ILD: riociguat increased hospitalizations and mortality in PH associated with fibrotic idiopathic interstitial pneumonia. By contrast, aerosolized/inhaled vasodilators act preferentially on ventilated lung and do not worsen shunt or oxygenation.

> *"Riociguat treatment increased hospitalizations and mortality."* — PMID: 33832831
>
> *"Systemic arterial pressure, arterial oxygen saturation, and pulmonary right-to-left-shunt flow, measured by multiple inert gas analysis, were not significantly changed."* — PMID: 10783653

Interpretation (ours): The inhaled route of ensifentrine plausibly converts a hazardous systemic vasodilator effect into a tolerable, ventilation-matched one — mirroring sildenafil's favorable V/Q behavior in lung fibrosis (PMID: 12354470). This tempers, but does not erase, the safety concern.

Finding 8 — No ensifentrine fibrosis data exist, and the mechanism can only slow, not reverse, IPF

PubMed searches for ensifentrine/RPL554 antifibrotic, fibroblast, myofibroblast, or collagen effects returned zero papers — there is no mechanistic fibrosis data for ensifentrine itself. Separately, even validated antifibrotics only slow decline: the PROGRESSION-IPF design notes that pirfenidone and nintedanib slow lung-function decline and uses FVC slope as the primary endpoint.

> *"antifibrotic therapies such as pirfenidone and nintedanib slow lung function decline"* — PMID: 42025256

Interpretation (ours): This defines a disease-stage ceiling. The plausible benefit of a PDE3/4 inhibitor is bounded by what the antifibrotic mechanism can do — slow progression, measured by FVC slope — not reverse established scar.

Finding 9 — The inhaled route is a genuine strategic advantage for tolerability

Oral pan-PDE4 inhibitor roflumilast is limited by class GI toxicity (nausea, diarrhea, weight loss, abdominal pain) causing substantial discontinuation; this prompted development of inhaled PDE4 inhibitors specifically to reduce systemic exposure while maximizing lung effect.

> *"This has prompted the search for PDE4 inhibitors to be given by inhalation to reduce the systemic exposure (and thus optimize the systemic safety) and maximize the therapeutic effect in the lung."* — PMID: 34887752

Inhaled CHF6001 (tanimilast) produced strong *local* anti-inflammatory gene modulation in sputum with no significant blood effect (PMID: 32197620). Notably, nerandomilast's most frequent AE in IPF was also GI (diarrhea) — exactly the toxicity an inhaled agent might reduce.

Interpretation (ours): This is the clearest advantage ensifentrine's route confers over oral nerandomilast — provided the drug reaches the diseased tissue (see Finding 6, the key tension).

Mechanistic Model / Interpretation

The hypothesis rests on a chain of inference. Each link can be assessed:

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IPF disease driver          Effector pathway          Drug
(genetics)                   (cAMP/PDE)               (ensifentrine)
─────────────────           ──────────────────        ──────────────────
MUC5B, telomeres,           PDE4B ↓ → cAMP ↑ →        Inhaled dual
surfactant, senescence  ──▶ ↓fibroblast activation ──▶ PDE3/4 inhibition
                            ↓myofibroblast diff.
[NOT the PDE pathway]       ↓profibrotic inflamm.     [ZERO fibrosis data]
     (Finding 5)            [VALIDATED by             [COPD only]
                             nerandomilast, F1/F3]     (Finding 2, 8)
                                   │
                            + PDE3 ↓ → cAMP ↑         Inhaled route:
                            (independent antifibrotic) + tolerability (F9)
                                  (Finding 4)         − tissue penetration (F6)
                                                      ± vasodilation (F7)
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The pathway link (middle column) is solid — nerandomilast proved cAMP elevation via PDE inhibition slows IPF, and both PDE3 and PDE4 arms feed the same antifibrotic cAMP node. The upstream link (genetics) is *absent* — the pathway is an effector, not a cause, which bounds the drug to disease-slowing. The downstream link (ensifentrine specifically) is *entirely unpopulated by data* — every statement about ensifentrine in fibrosis is extrapolation.

The decisive uncertainties cluster at the drug level:

Factor	Direction	Strength of evidence
Pathway active in IPF tissue	**For**	Strong (F3, review + preclinical)
Pathway clinically validated (nerandomilast)	**For**	Strong (F1, Phase 3)
PDE3 adds independent antifibrotic node	**For**	Moderate (F4, cilostazol preclinical)
Inhaled route → better tolerability	**For**	Moderate (F9, class rationale)
Pathway genetically causal	**Against**	Strong (F5, no GWAS signal)
Ensifentrine fibrosis data	**Against (absence)**	Strong (F2, F8, zero studies)
Inhaled penetration of fibrotic lung	**Against**	Moderate (F6, delivery barrier)
PDE3 vasodilation safety	**Mixed**	Moderate (F7, systemic harm vs inhaled safety)
Mechanism reverses fibrosis	**Against (ceiling)**	Strong (F8, only slows)

Ensifentrine vs. nerandomilast — the core comparison

Attribute	Nerandomilast (validated)	Ensifentrine (hypothesized)
Target	PDE4B-preferential	Dual PDE3 + PDE4
Route	Oral	Inhaled (nebulized)
IPF evidence	Phase 3 positive (P 40387033)	None
Fibrosis data	Preclinical + Phase 3	None at any level
Reaches fibrotic lesion via	Systemic circulation	Aerosol deposition (barrier, F6)
Main toxicity	GI (diarrhea)	COPD data: generally well tolerated
Added mechanism	—	PDE3 vasodilation + extra cAMP node

The comparison makes the hypothesis's logic explicit: ensifentrine *could* match or exceed nerandomilast if (a) dual PDE3/4 inhibition elevates cAMP more fully in fibroblasts, and (b) inhaled delivery achieves adequate tissue concentration. Neither has been demonstrated.

Evidence Base

PMID	Role	How it bears on the hypothesis
40387033	**Supports pathway**	Phase 3 FIBRONEER-IPF: PDE4B inhibition slows FVC decline (68.8 mL, P<0.001) — validates the target class
40331465	**Context/against**	Confirms ensifentrine's approved indication is COPD, not IPF
42768757	**Supports mechanism**	PDE4/cAMP pathway acts across IPF effector cells
41771338	**Supports mechanism**	PDE4 inhibitor matches nintedanib's antifibrotic effect in bleomycin mice
42207281	**Supports PDE3 arm**	PDE3 flagged as antifibrotic node; cilostazol validated in bleomycin rats
31884570	**Supports mechanism**	cAMP elevation is intrinsically antifibrotic in lung fibroblasts
25837031	**Against (genetics)**	IPF genetics center on surfactant/telomere biology, not cAMP/PDE
31034279	**Against (genetics)**	Confirms host-defense/senescence genes drive IPF risk
42359544	**Against (delivery)**	Fibrotic mucus/ECM physically obstruct inhaled drug penetration
37843890	**Against (delivery)**	Heterogeneous aerosol distribution in fibrotic lung
33832831	**Safety flag**	Systemic vasodilator (riociguat) increased mortality in fibrotic ILD
10783653	**Mitigates safety flag**	Inhaled vasodilators preserve oxygenation/shunt
12354470	**Mitigates safety flag**	Sildenafil preferentially vasodilates and improves gas exchange in lung fibrosis
42025256	**Defines ceiling/ endpoint**	Antifibrotics slow (not reverse) decline; FVC slope is the endpoint
34887752	**Supports route**	Rationale that inhaled PDE4 inhibition improves systemic safety
32197620	**Supports route**	Inhaled CHF6001 modulates lung inflammation locally, not systemically

Supporting context from the broader literature: a dedicated review of antifibrotic PDE4 strategies in IPF frames nerandomilast as the lead and discusses optimized drug delivery as a translational challenge (PMID: 41726771); the mechanism of ensifentrine as a dual PDE3/4 bronchodilator/anti-inflammatory is reviewed for COPD (PMID: 39688360); and ensifentrine's anti-inflammatory and anti-oxidative effects have been shown in COPD-relevant cells but not in fibrosis models (PMID: 41237701).

Limitations and Knowledge Gaps

1. **This is a literature-based assessment, not a primary-data analysis.** No experimental dataset on ensifentrine in fibrosis exists to analyze; the conclusions rest entirely on the published record and mechanistic inference.
2. **The central gap is the complete absence of ensifentrine fibrosis data.** There is no in vitro fibroblast assay, no bleomycin model, and no human IPF/ILD exposure for ensifentrine. Every favorable inference is borrowed from (a) a different molecule in the class (nerandomilast, oral PDE4B) or (b) a different PDE3 inhibitor (cilostazol).
3. **No biomarker-defined subgroup is established.** The hypothesis cannot currently say which IPF patients (if any) would benefit; unlike MUC5B genotype for prognosis, there is no cAMP/PDE-pathway biomarker to enrich a trial.
4. **The inhaled-delivery barrier is characterized only indirectly** — via nanoparticle and model-compound studies in bleomycin mice, not via ensifentrine deposition data in human fibrotic lung.
5. **PDE3 vasodilation safety in IPF is inferred by analogy.** The reassuring inhaled-vasodilator data (prostacyclin, sildenafil) are not ensifentrine-specific, and the harmful systemic signal (riociguat) comes from a different drug class.
6. **The effect ceiling is a slowing of FVC decline, not reversal** — so even a fully successful program would deliver incremental benefit, likely on top of existing antifibrotics.

Proposed Follow-up Experiments / Actions

The investigation points to a clear, efficient sequence that prioritizes de-risking the drug-specific unknowns before committing to an expensive pivotal trial.

1. **Proof-of-mechanism in human IPF tissue (highest priority, most efficient single experiment).** Treat patient-derived IPF lung fibroblasts and **precision-cut lung slices (PCLS)** from explanted IPF lungs with ensifentrine, benchmarked head-to-head against nerandomilast and nintedanib. Primary readouts: intracellular cAMP, α-SMA / myofibroblast differentiation, collagen-I and fibronectin deposition, and TGF-β-driven gene signatures. This directly tests whether ensifentrine engages the validated antifibrotic node in the *correct human tissue* and whether its dual PDE3/4 action matches or exceeds PDE4B-preferential inhibition.
2. **Aerosol-deposition / tissue-penetration study.** Using fibrotic animal models and ex vivo human fibrotic lung, quantify whether nebulized ensifentrine achieves antifibrotic concentrations at the fibroblastic foci — the specific barrier identified in Finding 6. A negative result here would be a decisive, early stop.
3. **In vivo bleomycin model.** Standard preventive and therapeutic dosing of inhaled ensifentrine with Ashcroft score, hydroxyproline, and FVC-analog endpoints, compared to nintedanib — the minimum preclinical package any PDE inhibitor in this space has (DC591017, cilostazol) already cleared.
4. **Safety/mechanistic sub-study on pulmonary hemodynamics and V/Q** in a fibrotic model, to confirm the inhaled route neutralizes the PDE3 vasodilation hazard seen with systemic agents.
5. **Only if steps 1–4 are positive:** a Phase 2a trial in IPF. Design — randomized, placebo-controlled, add-on to standard antifibrotic, **≥52-week FVC-slope primary endpoint** (per the FIBRONEER/PROGRESSION-IPF template), with an embedded aerosol-deposition/ biomarker sub-study. Consider enrichment by features of active inflammation given the pathway's anti-inflammatory bias.

Verdict

How strong is the case? Moderate on the pathway, weak on the molecule. The cAMP/PDE pathway is a *bona fide*, clinically validated antifibrotic target in IPF (nerandomilast, Phase 3), it is mechanistically active in the right cells, and ensifentrine's dual PDE3/4 action plus inhaled tolerability give it a coherent, even attractive, pharmacological rationale. But the pathway is a downstream effector (not genetically causal), the benefit ceiling is disease-slowing rather than reversal, inhaled delivery into fibrotic tissue is a real and specific barrier, and — most importantly — **ensifentrine has no fibrosis data of any kind.** The hypothesis is therefore a legitimate, testable repurposing idea at an early, low-confidence stage, not a near-ready clinical bet. The single most efficient test is a **human IPF precision-cut-lung-slice / primary-fibroblast proof-of-mechanism study benchmarking ensifentrine against nerandomilast, paired with an aerosol-deposition check** — this would, at modest cost, establish whether the drug engages the validated node in the correct tissue and reaches it, before any ≥52-week FVC-endpoint trial is contemplated.

Highest-weight identifiers:

- **PMID: 40387033 / NCT05321069** — Phase 3 FIBRONEER-IPF: validates PDE4B inhibition slows FVC decline; the keystone for the pathway.
- **PMID: 40331465** — Establishes ensifentrine's evidence base is COPD, defining the extrapolation gap.
- **PMID: 42207281** — PDE3 (ensifentrine's second target) is an independent antifibrotic node in lung fibrosis.
- **PMID: 25837031** — IPF genetics do not implicate the PDE/cAMP pathway; it is an effector, not a driver.
- **PMID: 42359544** — The fibrotic lung physically obstructs inhaled drug delivery, the key drug-specific risk.
- **PMID: 33832831** — Systemic pulmonary vasodilators harm fibrotic ILD patients (riociguat mortality); the PDE3 safety flag.
- **PMID: 34887752** — Pharmacological rationale that inhaled PDE4 inhibition improves systemic safety; ensifentrine's route advantage.
- **NCT04535986 / NCT04542057 (ENHANCE-1/2)** — Ensifentrine's pivotal COPD trials; the sole human efficacy base.